

Supplementary Material

TRUST-BC: A Multicohort Benchmark for Leakage-Free, Calibrated, and Uncertainty-Aware Breast Cancer Prediction Models

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Page numbers in the checklists refer to the PDF version of the revised main manuscript (section numbers are identical in the Word version). All tables in this document are also provided as CSV files in the repository folder `supplementary/tables/`, and every number can be regenerated with the scripts in `scripts/`.

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S0. Contents of this supplement

- Table S1: completed TRIPOD+AI checklist; Table S1b: TRIPOD+AI for Abstracts checklist.
- Table S2: author self-assessment against PROBAST+AI (development and evaluation parts).
- Figure S1: participant flow diagram.
- Tables S3 and S3b: participant characteristics by partition and by race (SEER).
- Table S4: number of participants and events in each analysis.
- Table S5: Nadeau–Bengio corrected confidence intervals for all pipelines.
- Table S6: conformal coverage by outcome class and subgroup (SEER).
- Table S7: leakage and selector contrasts, all cohorts and label versions.
- Section S8: SEER label audit, v1.0 results, reproduction check.
- Table S9: minimum sample size (Riley criteria).

Table S1. TRIPOD+AI checklist

Item descriptions are abbreviated from the TRIPOD+AI statement (Collins et al., BMJ 2024;385:e078378). D = development, E = evaluation; this study reports both.

Item	Topic	Status	Location and notes
1	Title: development/evaluation, population, outcome	Partial	Title identifies a breast cancer prediction benchmark; the study type (development with internal validation) is stated in the Abstract and Sec. 1.3, p. 3.
2	Abstract	Reported	Abstract, p. 1; see Table S1b.
3a	Healthcare context; diagnostic or prognostic; existing models	Reported	Sec. 1.1, p. 1; Sec. 4.4, p. 21.
3b	Target population, intended purpose and users	Reported	Sec. 1.1, p. 1; Sec. 4.3, p. 21.
3c	Known health inequalities between groups	Reported	Sec. 1.2, p. 2 (final paragraph).
4	Study objectives, development and/or evaluation	Reported	Sec. 1.3, p. 3.
5a	Data sources and representativeness	Reported	Sec. 2.1, p. 4.
5b	Dates of accrual and end of follow-up	Reported	Sec. 2.1, p. 4. WBCD: before 1993; Coimbra: 2009–2013; SEER: diagnoses 2006–2010, follow-up to 107 months.
6a	Setting, number and location of centres	Reported	Sec. 2.1, p. 4.
6b	Eligibility criteria	Reported	Sec. 2.1, p. 4; Sec. 2.2, p. 6.
6c	Treatments and how handled	Reported	Sec. 2.1, p. 4 (paragraph after cohort list).
7	Data preprocessing and quality checks, across groups	Reported	Sec. 2.3, p. 6 (Data preparation).
8a	Outcome definition, time horizon, rationale	Reported	Sec. 2.2, p. 6; corrected SEER labelling, Section S8.
8b	Qualifications of outcome assessors	Reported	Sec. 2.2, p. 6 (histopathology; registry linkage).
8c	Blinding of outcome assessment	Reported	Sec. 2.2, p. 6.
9a	Choice of initial predictors and pre-selection	Reported	Sec. 2.1, p. 4 (Predictors).
9b	Predictor definitions, how and when measured	Reported	Sec. 2.1, p. 4.
9c	Qualifications of predictor assessors	Reported	Sec. 2.1, p. 4 (WBCD operator-initialised boundaries; SEER registrars).
10	Sample size justification	Reported	Sec. 2.1, p. 4 (Sample size); Table S9.
11	Missing data	Reported	Sec. 2.2, p. 6. No missing values; no imputation.
12a	How data were used (partitioning)	Reported	Sec. 2.2, p. 6; Sec. 2.3, p. 6; Figure 1.
12b	Handling of predictors	Reported	Sec. 2.1, p. 4; Sec. 2.3, p. 6.
12c	Model type, building, tuning, internal validation	Reported	Sec. 2.3, p. 6; Sec. 2.4, p. 7.
12d	Heterogeneity across clusters	Reported	Sec. 2.3, p. 6: not examinable (single-centre diagnostic data; no registry identifiers).
12e	Performance measures and model comparison	Reported	Sec. 2.5, p. 7; Sec. 2.7, p. 9.
12f	Model updating	Reported	Sec. 2.3, p. 6: none performed.
12g	How evaluation predictions were calculated	Reported	Sec. 2.3, p. 6; Sec. 2.6, p. 9.
13	Class imbalance	Reported	Sec. 2.3, p. 6 (Data preparation, class imbalance and output).
14	Fairness approaches	Reported	Sec. 2.8, p. 9.
15	Model output and thresholds	Reported	Sec. 2.3, p. 6.
16	Differences between development and evaluation data	Reported	Sec. 2.2, p. 6.
17	Ethics	Reported	Declarations, p. 23.
18a	Funding	Reported	Declarations.
18b	Conflicts of interest	Reported	Declarations.
18c	Protocol	Reported	Declarations: no formal protocol; analysis plan fixed in code.
18d	Registration	Reported	Declarations: not registered.
18e	Data sharing	Reported	Declarations.
18f	Code sharing	Reported	Declarations (release v1.1, MIT licence).

Item	Topic	Status	Location and notes
19	Patient and public involvement	Reported	Declarations: none.
20a	Participant flow	Reported	Sec. 3.1, p. 10; Figure S1.
20b	Participant characteristics, by group	Reported	Tables S3 and S3b.
20c	Development vs evaluation distributions	Reported	Table S3 (standardised mean differences).
21	Participants and events per analysis	Reported	Table S4.
22	Full model specification / access	Reported	Sec. 2.3, p. 6; code and fitted pipelines in repository.
23a	Performance with CIs, including subgroups	Reported	Tables 2–5 of the main text; Tables S5–S7.
23b	Heterogeneity across clusters	N/A	Not examined (item 12d).
24	Model updating results	N/A	No updating.
25	Interpretation, including fairness	Reported	Sec. 4, p. 19; Sec. 3.6, p. 17.
26	Limitations	Reported	Sec. 4.5, p. 21.
27a	Handling of poor-quality or missing input	Reported	Sec. 4.3, p. 21 (Input data and user expertise).
27b	User interaction and expertise	Reported	Sec. 4.3, p. 21.
27c	Future research and generalisability	Reported	Sec. 4.5, p. 21; Sec. 5, p. 23; Sec. 6, p. 23.

Table S1b. TRIPOD+AI for Abstracts checklist

Item	Topic	Status	Location in abstract
1	Title	Reported	Title.
2	Background	Reported	Background.
3	Objectives	Partial	Background and Methods (no separate objective heading).
4	Sources of data	Reported	Methods.
5	Eligibility criteria and setting	Partial	Methods (cohorts and sizes); details in Section 2.1.
6	Outcome and time horizon	Partial	Methods (5-year classification cohort); outcome definitions in Section 2.2.
7	Model type, building and internal validation	Reported	Methods.
8	Performance measures	Reported	Methods.
9	Number of participants and events	Reported	Methods.
10	Predictors in final model	Partial	Methods (PCA or random-forest feature selection); predictors in Section 2.1.
11	Performance estimates with CIs	Reported	Results.
12	Interpretation	Reported	Conclusions.
13	Registration	Not in abstract	Stated in Declarations (not registered).

Table S2. PROBAST+AI author self-assessment

PROBAST+AI (Moons et al., BMJ 2025;388:e082505) has a model development part (concerns for *quality*) and a model evaluation part (*risk of bias*), each with four domains; applicability is rated for domains 1–3. This is an author self-assessment, not an independent appraisal. Ratings: Low, High, Unclear.

S2a. Model development — concerns for quality

Domain	WBCD	Coimbra	SEER 5-y	Rationale
1. Participants and data sources	Unclear	High	Unclear	WBCD: single centre; accrual details limited. Coimbra: case-control design with healthy controls. SEER: population-based but secondary public extraction, not re-derived from SEER*Stat; 754 patients with unknown 5-year status excluded.
2. Predictors	Low	Low	Low	Predictors available at the time of prediction; survival time and status excluded; preprocessing fitted within training folds. WBCD boundary initialisation is operator-dependent.
3. Outcome	Low	Low	Unclear	WBCD/Coimbra: pathology-based diagnosis. SEER: all-cause mortality from registry linkage; fixed-horizon labels corrected in v1.1, but exclusion of censored patients may bias the outcome distribution if censoring is informative.
4. Analysis	Low	High	Low	Nested CV; inner-fold preprocessing; calibration, discrimination and uncertainty reported. WBCD meets the Riley minimum (479 vs 483). Coimbra: EPV 5.8 and 98 vs 381 required. SEER: 2,779 vs 1,645 required; class weighting produced overprediction, which is reported.
Applicability (domains 1, 2, 3)	Low, Low, Low	High, Low, Low	Unclear, Low, Low	Coimbra case-control sample does not represent a screening population. SEER restricted to node-positive women aged 30–69, diagnosed 2006–2010, without treatment data.
Overall quality	Low	High	Unclear	

S2b. Model evaluation (held-out partitions) — risk of bias

Domain	WBCD lockbox	Coimbra lockbox	SEER lockbox	SEER survival test	Rationale
1. Participants	Low	High	Unclear	Low	Random stratified split from the same source. Coimbra inherits the case-control design; SEER classification lockbox inherits the censoring exclusion.
2. Predictors	Low	Low	Low	Low	Same definitions; no lockbox information used in development.
3. Outcome	Low	Low	Unclear	Low	As in development.
4. Analysis	High	High	High	Low	Events: 32, 10, 69 (below ~100 recommended); WBCD calibration unstable (separation). Survival test: 123 events; censoring handled by the model; C-index CI reported.
Applicability (domains 1, 2, 3)	Low, Low, Low	High, Low, Low	Unclear, Low, Low	Unclear, Low, Low	Internal validation only.
Overall risk of bias	High	High	High	Low	Driven mainly by the number of events and, for Coimbra, the study design.

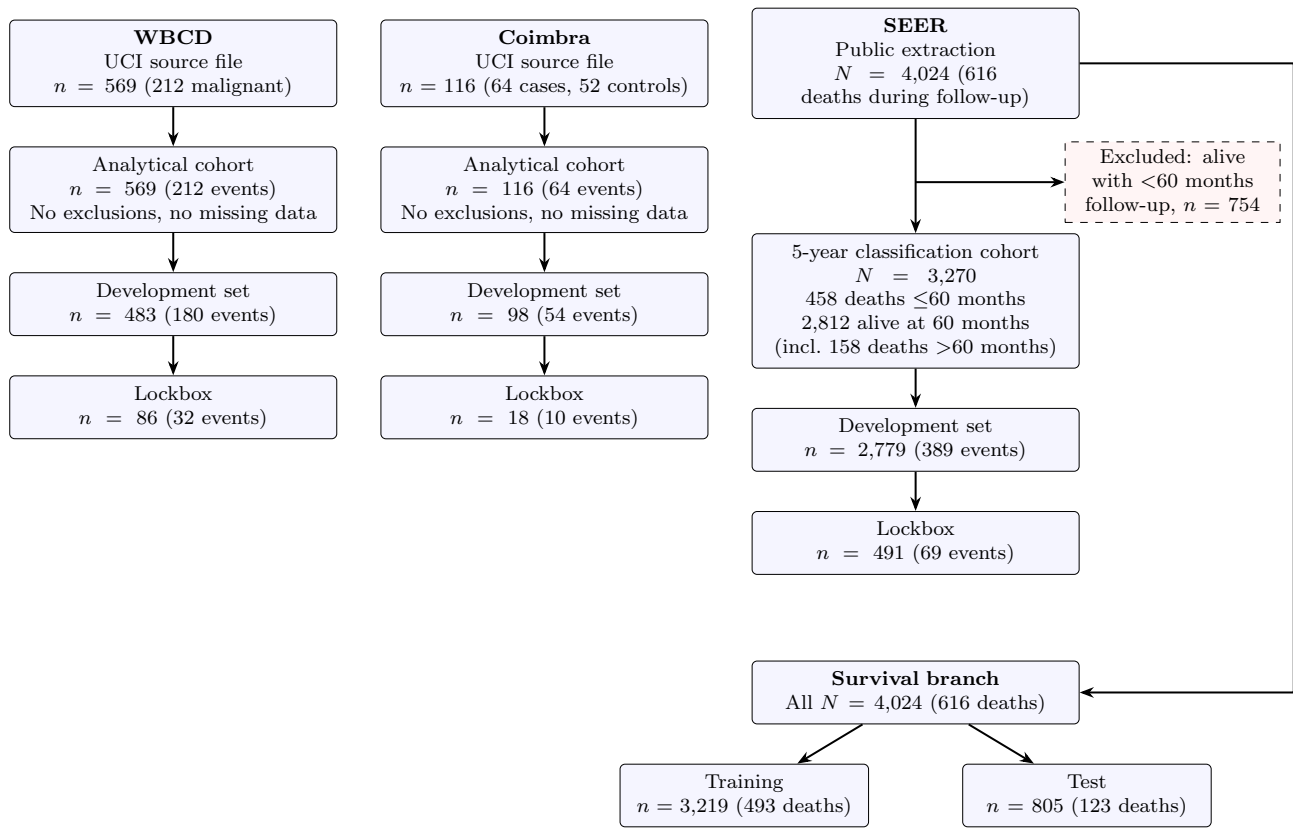
Figure S1. Participant flow**Figure S1:** Participant flow for the three cohorts and the SEER survival branch. Events are malignancy (WBCD), breast cancer (Coimbra), death from any cause within 60 months (SEER classification) and death from any cause during follow-up (survival branch).

Table S3. Participant characteristics by partition

SMD: standardised mean difference between development and lockbox (continuous variables). Characteristics by case/control status for Coimbra are reported in the source publication (Patrício et al., BMC Cancer 2018).

Characteristic	Development	Lockbox	SMD
WBCD (development $n=483$; lockbox $n=86$)			
mean radius, mean (SD)	14.2 (3.5)	13.9 (3.4)	0.06
mean texture, mean (SD)	19.4 (4.3)	18.9 (4.2)	0.12
mean perimeter, mean (SD)	92.2 (24)	90.8 (23)	0.06
mean area, mean (SD)	658 (3.6e+02)	636 (3.3e+02)	0.06
mean smoothness, mean (SD)	0.0961 (0.014)	0.0977 (0.013)	-0.11
mean compactness, mean (SD)	0.104 (0.053)	0.107 (0.05)	-0.05
mean concavity, mean (SD)	0.0888 (0.081)	0.0886 (0.074)	0.0
mean concave points, mean (SD)	0.0489 (0.039)	0.049 (0.036)	-0.0
mean symmetry, mean (SD)	0.181 (0.027)	0.18 (0.028)	0.04
mean fractal dimension, mean (SD)	0.0627 (0.0069)	0.0632 (0.0079)	-0.06
Malignant, n (%)	180 (37.3)	32 (37.2)	
Coimbra (development $n=98$; lockbox $n=18$)			
Age, mean (SD)	57.9 (16.4)	54.1 (14.1)	0.25
BMI, mean (SD)	27.7 (5.12)	27 (4.52)	0.13
Glucose, mean (SD)	98.9 (23.8)	91.7 (11.9)	0.38
Insulin, mean (SD)	10 (9.47)	10 (13.2)	-0.0
HOMA, mean (SD)	2.75 (3.69)	2.41 (3.48)	0.09
Leptin, mean (SD)	26.9 (19.5)	25.3 (17.7)	0.09
Adiponectin, mean (SD)	10.4 (7.18)	9.17 (4.6)	0.2
Resistin, mean (SD)	14.6 (12.6)	15.4 (11.3)	-0.06
MCP-1, mean (SD)	543 (344)	488 (363)	0.15
Breast cancer cases, n (%)	54 (55.1)	10 (55.6)	
SEER (development $n=2,779$; lockbox $n=491$)			
Age, years, mean (SD)	54.2 (9.0)	53.5 (8.9)	0.08
Race: Black, n (%)	209 (7.5)	34 (6.9)	
Race: Other (AI/AN, API), n (%)	209 (7.5)	46 (9.4)	
Race: White, n (%)	2361 (85.0)	411 (83.7)	
Marital status: Divorced, n (%)	346 (12.5)	67 (13.6)	
Marital status: Married (including common law), n (%)	1816 (65.3)	311 (63.3)	
Marital status: Separated, n (%)	35 (1.3)	3 (0.6)	
Marital status: Single (never married), n (%)	420 (15.1)	83 (16.9)	
Marital status: Widowed, n (%)	162 (5.8)	27 (5.5)	
T stage: T1, n (%)	1099 (39.5)	194 (39.5)	
T stage: T2, n (%)	1234 (44.4)	213 (43.4)	
T stage: T3, n (%)	366 (13.2)	74 (15.1)	
T stage: T4, n (%)	80 (2.9)	10 (2.0)	
N stage: N1, n (%)	1850 (66.6)	329 (67.0)	
N stage: N2, n (%)	581 (20.9)	95 (19.3)	
N stage: N3, n (%)	348 (12.5)	67 (13.6)	
Grade: Moderately differentiated; Grade II, n (%)	1630 (58.7)	272 (55.4)	
Grade: Poorly differentiated; Grade III, n (%)	776 (27.9)	155 (31.6)	
Grade: Undifferentiated; anaplastic; Grade IV, n (%)	15 (0.5)	3 (0.6)	
Grade: Well differentiated; Grade I, n (%)	358 (12.9)	61 (12.4)	
A stage: Distant, n (%)	64 (2.3)	13 (2.6)	
A stage: Regional, n (%)	2715 (97.7)	478 (97.4)	
Tumour size, mm, median (IQR)	25 (16-40)	25 (16-37)	-0.01
Estrogen receptor: Negative, n (%)	201 (7.2)	45 (9.2)	
Estrogen receptor: Positive, n (%)	2578 (92.8)	446 (90.8)	
Progesterone receptor: Negative, n (%)	516 (18.6)	98 (20.0)	
Progesterone receptor: Positive, n (%)	2263 (81.4)	393 (80.0)	
Regional nodes examined, median (IQR)	14 (9-19)	14 (9-19)	0.01
Regional nodes positive, median (IQR)	2 (1-5)	2 (1-5)	0.0
Follow-up, months, median (IQR)	79 (65-93)	79 (67-93)	-0.03
Death within 60 months, n (%)	389 (14.0)	69 (14.1)	
Missing values, n	0	0	

Table S3b. SEER classification cohort by race

Race	<i>n</i>	Age, mean	T3–T4 (%)	N3 (%)	ER negative (%)	Death ≤ 60 mo (%)
Black	243	52.7	14.4	17.3	12.8	23.9
Other	255	51.6	13.7	10.2	9.4	8.2
White	2,772	54.4	16.6	12.5	6.9	13.7

Table S4. Participants and events in each analysis

Analysis set	WBCD	Coimbra	SEER 5-year
Analytical cohort	569 (212)	116 (64)	3,270 (458)
Development set	483 (180)	98 (54)	2,779 (389)
Outer CV training fold (approx.)	386 (144)	78 (43)	2,223 (311)
Outer CV test fold (approx.)	97 (36)	20 (11)	556 (78)
Inner tuning folds (approx., training / validation)	257 / 129	52 / 26	1,482 / 741
Conformal: proper training / calibration / test	289 / 97 (36) / 97	58 / 20 (11) / 20	1,667 / 556 (78) / 556
Lockbox model: training / conformal calibration	386 (144) / 97 (36)	78 (43) / 20 (11)	2,223 (311) / 556 (78)
Lockbox	86 (32)	18 (10)	491 (69)
SEER survival: training / test	–	–	3,219 (493) / 805 (123)

Numbers are participants (events). Approximate values are for a typical fold of stratified splitting.

Table S5. Cross-validated performance with Nadeau–Bengio corrected 95% CIs

The v1.0 manuscript reported percentile-bootstrap intervals over outer-fold scores, which treat correlated folds as independent and are too narrow. The intervals below use the corrected variance $S^2(1/n + 1/(K - 1))$ with $K = 5$.

Cohort	Label	Pipeline	Metric	Mean	SD	95% CI (corrected)
WBCD	original	Leaky	Accuracy	96.85%	1.65%	95.13%–98.57%
WBCD	original	XGBoost	Accuracy	95.20%	2.08%	93.03%–97.37%
WBCD	original	LightGBM	Accuracy	95.82%	2.30%	93.41%–98.22%
WBCD	original	Nested_PCA	Accuracy	97.16%	1.65%	95.44%–98.88%
WBCD	original	Nested_RF	Accuracy	95.94%	1.94%	93.92%–97.97%
WBCD	original	Unified	Accuracy	97.08%	1.67%	95.34%–98.82%
Coimbra	original	Leaky	Accuracy	62.73%	10.70%	51.56%–73.91%
Coimbra	original	XGBoost	Accuracy	75.64%	9.72%	65.49%–85.79%
Coimbra	original	LightGBM	Accuracy	74.78%	10.66%	63.65%–85.92%
Coimbra	original	Nested_PCA	Accuracy	63.25%	10.88%	51.89%–74.61%
Coimbra	original	Nested_RF	Accuracy	72.37%	8.20%	63.81%–80.93%
Coimbra	original	Unified	Accuracy	71.35%	8.70%	62.27%–80.43%
SEER	v1.0 (as reported)	Leaky	AUROC	0.718	0.024	0.692–0.744
SEER	v1.0 (as reported)	XGBoost	AUROC	0.732	0.027	0.703–0.761
SEER	v1.0 (as reported)	LightGBM	AUROC	0.731	0.027	0.702–0.760
SEER	v1.0 (as reported)	Nested_PCA	AUROC	0.718	0.025	0.691–0.744
SEER	v1.0 (as reported)	Nested_RF	AUROC	0.732	0.023	0.707–0.757
SEER	v1.0 (as reported)	Unified	AUROC	0.732	0.023	0.707–0.757
SEER	v1.1 (fixed 60-month)	Leaky	AUROC	0.735	0.031	0.701–0.768
SEER	v1.1 (fixed 60-month)	Nested_PCA	AUROC	0.735	0.031	0.701–0.768
SEER	v1.1 (fixed 60-month)	Nested_RF	AUROC	0.741	0.027	0.713–0.770
SEER	v1.1 (fixed 60-month)	Unified	AUROC	0.739	0.028	0.709–0.769

Table S6. Conformal coverage by outcome class and subgroup (SEER, 100 iterations)

Variable	Group	Coverage (%)	Coverage in deaths (%)	Coverage in survivors (%)	Ambiguous sets (%)
race	Black	94.0	79.6	98.6	47.6
race	Other	96.5	62.2	99.7	33.7
race	White	95.3	68.8	99.4	37.6
age	<50	95.6	65.4	99.7	29.3
age	50-59	96.0	74.2	99.3	39.2
age	≥60	94.1	69.7	99.2	45.8
outcome	event	69.9	–	–	–
outcome	non-event	99.4	–	–	–

Coverage is marginal by construction; the large difference between deaths and survivors motivates class-conditional (Mondrian) conformal prediction in future work.

Table S7. Leakage and selector contrasts

Cohort	Label	Contrast	Mean difference	95% CI	p (two-sided)	p_{ToST}
WBCD	original	Unified-Leaky	+0.23 pts	−0.71 to +1.16	0.6251	0.0002
WBCD	original	Nested PCA-Leaky	+0.31 pts	−0.54 to +1.16	0.4669	0.0001
Coimbra	original	Unified-Leaky	+8.62 pts	−1.35 to +18.59	0.0886	0.9059
Coimbra	original	Nested PCA-Leaky	+0.52 pts	−4.68 to +5.72	0.8413	0.2851
SEER	v1.0	Unified-Leaky	+0.0140	−0.0006 to +0.0286	0.0588	0.7116
SEER	v1.0	Nested PCA-Leaky	−0.0001	−0.0016 to +0.0013	0.8449	<0.0001
SEER	v1.1	Unified-Leaky	+0.0043	−0.0107 to +0.0194	0.5607	0.2251
SEER	v1.1	Nested PCA-Leaky	+0.0000	−0.0014 to +0.0014	0.9583	<0.0001

Nested PCA – Leaky isolates the effect of fitting scaling and PCA outside the folds (post hoc, descriptive).

Unified – Leaky combines leakage with adaptive selector choice. Margins: ± 0.02 accuracy, ± 0.01 AUROC. p -values unadjusted.

S8. SEER label audit, v1.0 results and reproduction check

Audit. In the v1.0 code (`src/trust_bc_multicohort_benchmark.py`), the SEER label was $y = (\text{Status} == \text{'dead'})$ after excluding patients alive with fewer than 60 months of follow-up. Of the 616 deaths, 458 occurred at or before month 60 and 158 occurred between months 61 and 102. These 158 patients were alive at the 5-year horizon but were labelled as 5-year deaths, so the v1.0 endpoint was “death at any time during follow-up among patients with at least 60 months of follow-up or an observed death” rather than 5-year mortality.

Correction. Version 1.1 labels an event only if death occurred at or before month 60. The analytical cohort size is unchanged ($N = 3,270$) but the number of events falls from 616 to 458 and the prevalence from 18.8% to 14.0%.

Reproduction check. The v1.1 re-analysis re-implemented the Leaky, Nested PCA, Nested RF and Unified pipelines with identical code, seeds and splits (`scripts/seer_revision_core.py`) in a newer software environment (scikit-learn 1.8.0). Run with the v1.0 label, it reproduced all 35 outer-fold AUROC and accuracy values of the four pipelines reported in `results/SEER/SEER_raw_cv_scores.csv` (maximum absolute difference 0 at the stored precision), and the Unified pipeline selected RF in 35/35 folds as in v1.0. The split manifest was also reproduced exactly (491 identical lockbox indices).

Table S8. SEER results under the v1.0 label (as originally reported)

Pipeline	AUROC mean ± SD	95% CI (corrected)	Holm p vs Leaky	Holm p vs Unified	Brier / ECE
Leaky baseline	0.718 ± 0.024	0.692–0.744	–	–	0.155 / 0.133
Tuned XGBoost	0.732 ± 0.027	0.703–0.761	0.5011	1.0000	0.140 / 0.022
Tuned LightGBM	0.731 ± 0.027	0.702–0.760	0.5011	1.0000	0.149 / 0.052
Nested PCA-only	0.718 ± 0.025	0.691–0.744	–	–	0.155 / 0.133
Nested RF-only	0.732 ± 0.023	0.707–0.757	–	–	0.152 / 0.128
Unified	0.732 ± 0.023	0.707–0.757	0.2939	–	0.152 / 0.128

v1.0 TOST (Unified vs Leaky): $p = 0.7116$. v1.0 lockbox: accuracy 80.04%, AUROC 0.731, sensitivity 34.78% (32/92), specificity 90.48% (361/399), conformal coverage 96.54%; v1.0 conformal (development): coverage 95.32%, singleton 48.62%, ambiguous 51.38%.

Effect of the correction. Discrimination was similar (Unified AUROC 0.732 → 0.739), but the proportion of ambiguous conformal sets fell from 51.38% to 38.08%, and the Unified selector chose RF in 31/35 rather than 35/35 folds. The v1.0 claim of a leakage effect in SEER (Unified vs Leaky, 0.732 vs 0.718) is explained by the selector: under both labels the architecture-matched contrast was essentially zero (Table S7).

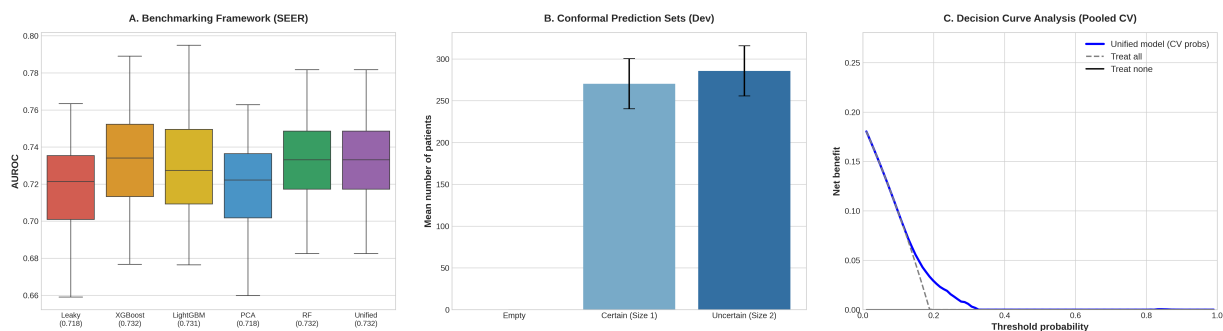


Figure S2: SEER results under the v1.0 label (original Figure 2c), retained for transparency.

Table S9. Minimum sample size for model development (Riley et al.)

Cox–Snell R^2 derived from the anticipated C statistic by simulation (Riley, Van Calster & Collins, Stat Med 2021). Criteria: (i) shrinkage ≥ 0.90 ; (ii) optimism in Nagelkerke $R^2 \leq 0.05$; (iii) overall risk within ± 0.05 .

Cohort	Parameters	C	Prevalence	R^2_{CS}	n (i) / (ii) / (iii)	Required	Available (dev)
WBCD	30	0.99	0.373	0.659	228 / 479 / 360	479	483
Coimbra	9	0.79	0.552	0.244	286 / 206 / 381	381	98
SEER 5-year	16	0.73	0.140	0.083	1,645 / 544 / 186	1,645	2,779

For evaluation, at least 100 events and 100 non-events are commonly recommended; the lockboxes contain 32, 10 and 69 events.